How Hybrid Cars Actually Work

Energy flows, power-split transaxles, braking recovery, and the control strategy behind a full hybrid vehicle.

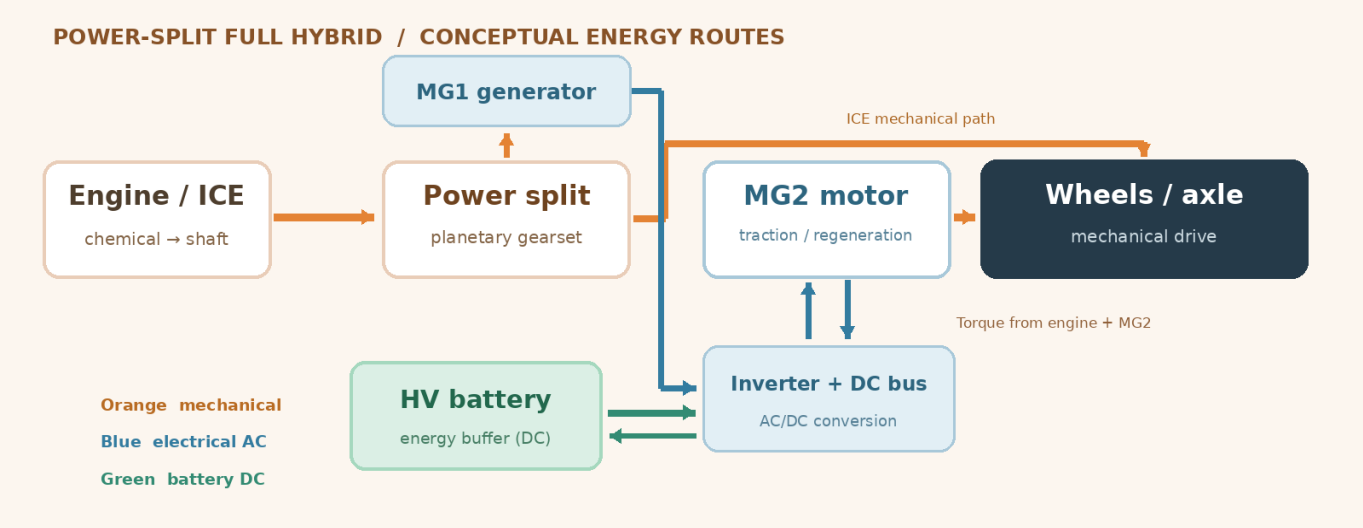


Figure 1. Simplified power-split topology; actual gearing and component layout vary by model and manufacturer.

1 Two power sources, one axle

Internal combustion engine (ICE). Fuel provides high energy density but engines lose efficiency in stop-start operation, idling, and low-load operation. The hybrid supervisory controller may stop the ICE or run it near favorable load points.

Electric side. The traction motor (MG2 in many power-split layouts) delivers wheel torque and can generate during braking. The high-voltage battery stores relatively limited energy; the inverter converts the battery's DC to motor AC and back.

2 The planetary power split

Series-parallel behavior. A planetary gearset couples the engine, generator (MG1), and driveline. Some engine power reaches the axle mechanically, while some drives a generator; electrical power can simultaneously feed MG2.

Continuously variable effect. MG1 speed / torque control changes the relationship between engine speed and road speed without a stepped conventional gearbox. Under hard acceleration, motor torque can supplement engine output; during coasting, engine-off operation may be possible.

Power balance (instantaneous) $P_{fuel} \approx P_{wheels} + P_{heat} + P_{accessories} + dE_{battery}/dt$ (idealized system boundary).

Operating modes and energy

TRADE-OFF MATRIX

Driving state	Engine	Electric machine	Battery trend
Launch / crawl	Usually off if conditions allow	MG2 supplies wheel torque	Discharging
Steady cruise	Often on near efficient load	Assist, generate or idle	Managed near target SOC
Hard acceleration	On, high output	MG2 adds torque	Often discharging
Brake / descend	Often fuel-cut or off	MG2 generates electricity	Charging if capacity permits
Parked / low SOC	May run for charging	MG1 can generate	Charging toward target
Cold / cabin heat	May run for thermal needs	Assist depends on controls	Model- and temperature-dependent

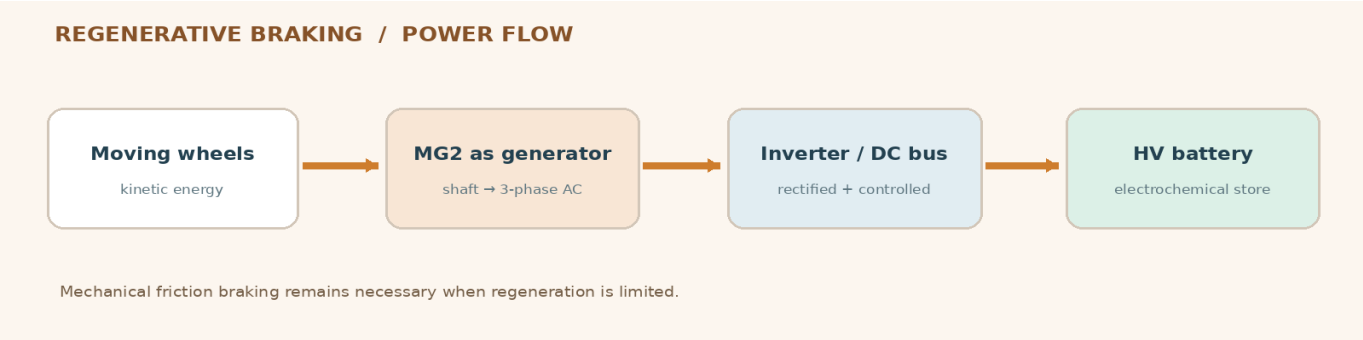


Figure 2. Braking changes the traction motor into a generator; energy still encounters conversion and storage losses.

Energy-recovery calculation

At 50 km/h, a 1,400 kg car carries **½mv² ≈ 135 kJ ≈ 0.0375 kWh** of translational kinetic energy. This is the theoretical ceiling before aerodynamic, rolling, electrical and battery losses.

Recovery is lower when the battery is full, cold or charge-limited, or when braking power exceeds the motor / battery acceptance capability. Friction brakes blend in for remaining deceleration.

Architecture comparison

Series: wheels are driven electrically; the engine turns a generator.
Parallel: engine and motor mechanically contribute wheel torque.
Power-split: planetary coupling provides mechanical and electrical paths.

Full hybrid vs PHEV: a full hybrid normally charges from the engine and regenerative braking. A plug-in hybrid carries a larger rechargeable battery and can take energy from the grid; actual EV range depends on vehicle design.

PRIMARY SOURCES

Toyota hybrid system Toyota technical review Prius ERG